Quiz 1

Graded

Student

Samyak Sanghvi

Total Points

4 / 4 pts

Question 1

(no title) **2** / 2 pts

- ✔ **+ 2 pts** Correct answer (200 pixels/cm, or 20 px/mm, or ~500 px/in)
- **− 0.5 pts** word "resolution" not mentioned
- **− 0.5 pts** result given in units of pixels/area
- **+ 0 pts** No/Wrong answer

Question 2

(no title) **2** / 2 pts

- ✔ **+ 0.5 pts** Correct: it is not linear
- ✔ **+ 1.5 pts** Valid counter example
- **+ 0 pts** No/Wrong answer

Name: Samyak COL7(6)83 Quiz 1 Entry no.: 2023CS10807

5 August, 2025

1. You are preparing a report and want to insert an image of size 1000×1000 pixels in a 5×5 cm region. What should be the specification of the printer, so that the image can be reproduced with no loss of information?

$$\frac{1000 \text{ pixel}}{5 \text{ cm}} = 200 \text{ p/cm}$$
$$\approx 400 \text{ dpi}$$

$\therefore$ printer must be of a $\approx$ 400dpi resolution.

2. Is intensity thresholding, as given below, a linear operation? Justify your answer with a proof or a counterexample.

$$T(r) = \begin{cases} 0 & \text{if } r < k, \\ L-1 & \text{if } r \geq k \end{cases}$$

$\boxed{\text{No}}$

$$T(k-1) = 0$$
$$T(2) = 0 \quad (\text{assuming } 2 < k)$$
$$T(k-1+2) = T(k+1) = L-1$$
$$\neq T(k-1) + T(2)$$

$$T(k-\delta) = 0 \quad (\cancel{\text{given } k > 2})$$

$\delta << K$

$$T(2k-2\delta) = L-1 \neq 2 \times T(k-\delta)$$

$$2k-2\delta > k$$